

NEXGO Android Smart PinPAD Programmer's Manual

Host Command Protocol Specification

Revision 1.2 - 2026-07-03

This document specifies the host command protocol implemented by the NEXGO Android Smart PinPAD application. The device performs local L2 card interaction, secure PIN entry, prompts, key management, and card-reader control. Host authorization and settlement remain outside the device.

Section 1 Communication Protocol

All command packets are transmitted through the selected serial transport. The application supports RS232 and USB CDC virtual serial operation. The transport layer validates start and end controls, LRC, partial-frame timeout, ACK/NAK retry, and session termination.

Control characters:

Packet control values		
Field	Length	Value and description
<STX>	1	02 hexadecimal. Starts a transaction frame.
<ETX>	1	03 hexadecimal. Ends a transaction frame.
<SI>	1	0F hexadecimal. Starts an administration frame.
<SO>	1	0E hexadecimal. Ends an administration frame.
<ACK>	1	06 hexadecimal. Positive acknowledgement.
<NAK>	1	15 hexadecimal. Transmission error acknowledgement.
<EOT>	1	04 hexadecimal. Session termination, format error, or cancellation.
<FS>	1	1C hexadecimal. Field separator.
<SUB>	1	1A hexadecimal. Subfield delimiter.

Message flow:

HOST	Dir.	PINPAD
Send request frame.	->	
	<-	Send <ACK> after a valid frame is received and accepted for processing. Send <NAK> on transmission error. Send <EOT> on packet format error where specified.
	<-	Send command response frame when processing completes.
Send <ACK> if response is valid, <NAK> if response has LRC error, or <EOT> to cancel.	->	
	<-	Administration responses send final <EOT> after host <ACK>, except commands 11 and 14. Transaction responses do not send a final <EOT>.

Section 2 Administration and Maintenance Messages

Message 02 Load Master Key or Working Key

Format: <SI>02[Data]<SO>[LRC] (request frame)
 <SI>02[Data]<SO>[LRC] (response frame)

Message length: Variable.

Usage: Loads a clear key into the selected secure key slot or working-key role.

Implementation: K0 writes the device master-key slot. P0, M3, and D0 are stored as working PIN, MAC, and data keys. P0 uses a random wrapper master key internally.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<SI>	1	<0F>
02	2	Message ID
[Key ID]	1	'0' to '9' and 'A' to 'G'.
[Key value]	16, 32, or 48	Hexadecimal key value. Single, double, and triple length keys are accepted according to the selected algorithm.
<FS>	1	Optional field separator. Present when key attributes follow.
[Usage]	2	Optional key usage. K0 key encryption, K1 key transport, P0 PIN encryption, M3 MAC, D0 data encryption.
[Mode]	1	Optional key mode. Common values include D, E, G, V.
[Algorithm]	1	Optional algorithm. Default is T when omitted.
<SO>	1	<0E>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<SI>	1	<0F>
02	2	Message ID
02	2	Message ID echoed with the original payload on success.
?	1	Present only in the error frame.
[Error code]	1	Command-specific key loading error code.
<SO>	1	<0E>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message 02 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request.

HOST	Dir.	PINPAD
		If successful, send Message 02. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	
	<-	Send final <EOT> after host <ACK>.

Message 04 Check Master Key

Format: <SI>04[Data]<SO>[LRC] (request frame)
 <SI>04[Data]<SO>[LRC] (response frame)

Message length: Variable.

Usage: Queries whether a key exists and optionally returns the stored key usage metadata.

Implementation: If secure key material exists but local metadata was removed by reinstall, PIN commands fail safely and require key injection again.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<SI>	1	<0F>
04	2	Message ID
[Key ID]	1	Key slot to query.
[Attribute flag]	1	Optional. '1' requests key attribute information.
<SO>	1	<0E>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<SI>	1	<0F>
04	2	Message ID
[Status]	1	'F' key exists, '0' key does not exist.
[Usage]	2	Returned only when requested and available.
<FS>	1	Field separator.
[Mode]	1	Returned key mode.
[Algorithm]	1	Returned algorithm.
<SO>	1	<0E>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
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HOST	Dir.	PINPAD
Message 04 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message 04. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	
	<-	Send final <EOT> after host <ACK>.

Message 05 Load Serial Number

Format: <SI>05[Data]<SO>[LRC] (request frame)
<SI>05[Data]<SO>[LRC] (response frame)

Message length: Variable.

Usage: Compatibility command. The hardware serial number is not changed.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<SI>	1	<0F>
05	2	Message ID
[Serial number]	0 to 16	Compatibility serial number value.
<SO>	1	<0E>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<SI>	1	<0F>
05	2	Message ID
[Serial number]	0 to 16	Echo of accepted serial substring.
<SO>	1	<0E>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message 05 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request.

HOST	Dir.	PINPAD
		If successful, send Message 05. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	
	<-	Send final <EOT> after host <ACK>.

Message 06 Get Serial Number

Format: <SI>06[Data]<SO>[LRC] (request frame)
 <SI>06[Data]<SO>[LRC] (response frame)

Message length: Fixed request.

Usage: Returns the device serial number.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<SI>	1	<0F>
06	2	Message ID
<SO>	1	<0E>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<SI>	1	<0F>
06	2	Message ID
[Serial number]	Variable	Device serial number from Android or NEXGO SDK.
<SO>	1	<0E>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message 06 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message 06. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>.	->	

HOST	Dir.	PINPAD
If canceling the session, send <EOT>.		
	<-	Send final <EOT> after host <ACK>.

Message 08 Select Active Master Key

Format: <SI>08[Data]<SO>[LRC] (request frame)
 <SI>08[Data]<SO>[LRC] (response frame)

Message length: Fixed.

Usage: Selects the active key used by MK/SK PIN and MAC commands.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<SI>	1	<0F>
08	2	Message ID
[Key ID]	1	Key id to select.
<SO>	1	<0E>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<SI>	1	<0F>
08	2	Message ID
[Status]	1	'0' success, '1' failure.
<SO>	1	<0E>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message 08 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message 08. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	
	<-	Send final <EOT> after host <ACK>.

Message 09 Communications Test

Format: <SI>09[Data]<SO>[LRC] (request frame)
 <SI>09[Data]<SO>[LRC] (response frame)

Message length: Fixed.

Usage: Performs a two-step communication loopback test.

Implementation: The host must echo the first response payload. The device then returns status 0 for success.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<SI>	1	<0F>
09	2	Message ID
<SO>	1	<0E>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<SI>	1	<0F>
09	2	Message ID
<SUB>	1	Subfield delimiter.
PROCESSING	10	Processing echo text, followed by final status in the second exchange.
<SO>	1	<0E>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message 09 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message 09. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	
	<-	Send final <EOT> after host <ACK>.

Message 11 Device Connection Test

Format: <SI>11[Data]<SO>[LRC] (request frame)
 <ACK>/<NAK>/<EOT> only unless an error frame is specified.

Message length: Fixed.

Usage: Connection probe. A valid frame is acknowledged without a final EOT.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<SI>	1	<0F>
11	2	Message ID
<SO>	1	<0E>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<ACK>	1	Command accepted
<NAK>	1	Transmission error
<EOT>	1	Format error, timeout, or canceled session

Message flow:

HOST	Dir.	PINPAD
Message 11 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Apply command action. No data frame is sent for a normal accepted command.

Message 12 Select Prompt Language

Format: <SI>12[Data]<SO>[LRC] (request frame)
 <SI>12[Data]<SO>[LRC] (response frame)

Message length: Fixed.

Usage: Compatibility prompt-language command.

Implementation: The Android application language follows Android locale settings.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<SI>	1	<0F>
12	2	Message ID
[Language index]	1	Language table index requested by host.
<SO>	1	<0E>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description

<SI>	1	<0F>
12	2	Message ID
[Status]	1	'0' accepted, '1' rejected.
<SO>	1	<0E>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message 12 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message 12. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	
	<-	Send final <EOT> after host <ACK>.

Message 13 Set Communication Speed

Format: <SI>13[Data]<SO>[LRC] (request frame)
 <SI>13[Data]<SO>[LRC] (response frame)

Message length: Fixed.

Usage: Validates a communication speed request.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<SI>	1	<0F>
13	2	Message ID
[Baud code]	1	Baud-rate selector 1 through 8.
[Mode]	0 or 1	Optional communication mode selector.
<SO>	1	<0E>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<SI>	1	<0F>
13	2	Message ID
[Status]	1	'0' accepted, '1' rejected.

<SO>	1	<0E>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message 13 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message 13. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	
	<-	Send final <EOT> after host <ACK>.

Message 14 Acknowledge Without Final End

Format: <SI>14[Data]<SO>[LRC] (request frame)
<ACK>/<NAK>/<EOT> only unless an error frame is specified.

Message length: Fixed.

Usage: Compatibility acknowledgement command.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<SI>	1	<0F>
14	2	Message ID
<SO>	1	<0E>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<ACK>	1	Command accepted
<NAK>	1	Transmission error
<EOT>	1	Format error, timeout, or canceled session

Message flow:

HOST	Dir.	PINPAD
Message 14 request frame	->	
	<-	<ACK>/<NAK>/<EOT>

HOST	Dir.	PINPAD
	<-	Apply command action. No data frame is sent for a normal accepted command.

Message 17 Request Random Number

Format: <SI>17[Data]<SO>[LRC] (request frame)
 <SI>17[Data]<SO>[LRC] (response frame)

Message length: Fixed.

Usage: Returns an 8-byte random challenge.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<SI>	1	<0F>
17	2	Message ID
<SO>	1	<0E>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<SI>	1	<0F>
17	2	Message ID
[Random]	16	Eight random bytes encoded as hexadecimal.
<SO>	1	<0E>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message 17 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message 17. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	
	<-	Send final <EOT> after host <ACK>.

Message 19 Query Firmware Version

Format: <SI>19[Data]<SO>[LRC] (request frame)
 <SI>19[Data]<SO>[LRC] (response frame)

Message length: Variable.

Usage: Returns component firmware or application version.

Implementation: The version string is formatted for compatibility and includes the NEXGO model when available.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<SI>	1	<0F>
19	2	Message ID
[Component]	1	Firmware component selector.
[Option]	0 or 1	Optional component option.
<SO>	1	<0E>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<SI>	1	<0F>
19	2	Message ID
.	1	Version prefix separator.
[Version]	Variable	Application, SDK, or device firmware version string.
<SO>	1	<0E>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message 19 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message 19. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	
	<-	Send final <EOT> after host <ACK>.

Message 1C Query Hardware Capability

Format: <SI>1C[Data]<SO>[LRC] (request frame)
<SI>1C[Data]<SO>[LRC] (response frame)

Message length: Variable.

Usage: Returns supported hardware and application capability values.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<SI>	1	<0F>
1C	2	Message ID
<SO>	1	<0E>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<SI>	1	<0F>
1C	2	Message ID
<FS>	1	Field separator before each capability item.
[Capability]	Variable	Capability string.
<SO>	1	<0E>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message 1C request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message 1C. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	
	<-	Send final <EOT> after host <ACK>.

Message 1F Query Prompt Table

Format: <SI>1F[Data]<SO>[LRC] (request frame)
<SI>1F[Data]<SO>[LRC] (response frame)

Message length: Variable.

Usage: Reports usable prompt tables.

Message element:

Request frame (HOST to PINPAD)

Field	Length	Value and description
<SI>	1	<0F>
1F	2	Message ID
<SO>	1	<0E>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<SI>	1	<0F>
1F	2	Message ID
[Language list]	Variable	Prompt-language identifiers separated by <FS>.
<SO>	1	<0E>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message 1F request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message 1F. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	
	<-	Send final <EOT> after host <ACK>.

Message 1M Setup Keypad Beeper

Format: <SI>1M[Data]<SO>[LRC] (request frame)
 <SI>1M[Data]<SO>[LRC] (response frame)

Message length: Fixed.

Usage: Controls keypad beep preference.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<SI>	1	<0F>
1M	2	Message ID
[Option]	1	'0' disable, '1' enable.
<SO>	1	<0E>

[LRC]	1	Checksum
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Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<SI>	1	<0F>
1M	2	Message ID
[Status]	1	Echoed option on success. '0' on invalid request.
<SO>	1	<0E>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message 1M request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message 1M. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	
	<-	Send final <EOT> after host <ACK>.

Message 1P Control Beeper

Format: <SI>1P[Data]<SO>[LRC] (request frame)
 <SI>1P[Data]<SO>[LRC] (response frame)

Message length: Variable.

Usage: Plays one or more beeps.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<SI>	1	<0F>
1P	2	Message ID
<SUB>	1	Subfield delimiter.
[Count]	1	Number of beeps, 1 through 9.
<FS>	1	Field separator.
[Duration]	1 to 3	Duration in 10 ms units.
[Interval]	1 to 3	Interval in 10 ms units.
<SO>	1	<0E>

[LRC]	1	Checksum
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Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<SI>	1	<0F>
1P	2	Message ID
[Status]	1	'0' success, '1' failure.
<SO>	1	<0E>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message 1P request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message 1P. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	
	<-	Send final <EOT> after host <ACK>.

Message M03 Load Permanent Unit Serial Number

Format: <SI>M03[Data]<SO>[LRC] (request frame)
<SI>M03[Data]<SO>[LRC] (response frame)

Message length: Variable.

Usage: Stores a reinstall-persistent private serial number.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<SI>	1	<0F>
M03	3	Message ID
[Serial]	11	Uppercase alphanumeric or dash serial value.
[Slot]	0 or 1	Optional slot 0 through 7. Default 0.
<SO>	1	<0E>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)

Field	Length	Value and description
<SI>	1	<0F>
M03	3	Message ID
[Status]	1	'0' success, '1' bad serial, '2' already set, '3' storage or slot error.
<SO>	1	<0E>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message M03 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message M03. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	
	<-	Send final <EOT> after host <ACK>.

Message M04 Query Permanent Unit Serial Number

Format: <SI>M04[Data]<SO>[LRC] (request frame)
<SI>M04[Data]<SO>[LRC] (response frame)

Message length: Variable.

Usage: Returns the permanent unit serial number.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<SI>	1	<0F>
M04	3	Message ID
[Slot]	0 or 1	Optional slot. Default 0.
<SO>	1	<0E>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<SI>	1	<0F>
M04	3	Message ID
[Serial]	0 or 11	Stored permanent serial, empty if none was loaded.

<SO>	1	<0E>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message M04 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message M04. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	
	<-	Send final <EOT> after host <ACK>.

Section 3 Security and PIN Messages

Message 20 Load Secret Master Key

Format: <STX>20[Data]<ETX>[LRC] (request frame)
 <STX>20[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Loads the secret master key used by secret PIN session slots.

Implementation: The key is injected into internal secure slots 20 through 29. Session-key loading waits until this master key is ready.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
20	2	Message ID
[Option]	1	'0' or '1'.
[Key value]	16 or 32	Secret master key in hexadecimal.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
20	2	Message ID
20	2	Message ID echoed with original payload on success.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message 20 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message 20. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message 21 Load Secret Session Key

Format: <STX>21[Data]<ETX>[LRC] (request frame)

<STX>21[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Loads a secret MK/SK session key for PIN entry.

Implementation: Key id 0 through 9 maps internally to secure slots 20 through 29.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
21	2	Message ID
[Key ID]	1	Secret session key id 0 through 9.
[Key value]	16 or 32	Encrypted PIN session key in hexadecimal.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
21	2	Message ID
21	2	Message ID echoed with original payload on success.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message 21 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message 21. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message 22 PIN Entry Request Using Secret MK/SK

Format: <STX>22[Data]<ETX>[LRC] (request frame)

<STX>71[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Starts secure online PIN entry using the secret key family.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
22	2	Message ID
[Account]	8 to 19	PAN/account digits used for PIN block formatting.
<FS>	1	Field separator.
[Key/control]	Variable	Secret key id followed by optional PIN control and prompt data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
71	2	Message ID
[Status/PIN data]	Variable	Message 71 payload containing PIN block or error code.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message 22 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Accept valid request and start the operation. Send the response when the operation completes. If successful, send Message 71. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message 23 PIN Entry Request With External Prompt

Format: <STX>23[Data]<ETX>[LRC] (request frame)
 <STX>71[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Starts secure online PIN entry using the secret key family.

Message element:

Request frame (HOST to PINPAD)

Field	Length	Value and description
<STX>	1	<02>
23	2	Message ID
[Account]	8 to 19	PAN/account digits used for PIN block formatting.
<FS>	1	Field separator.
[Key/control]	Variable	Secret key id followed by optional PIN control and prompt data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
71	2	Message ID
[Status/PIN data]	Variable	Message 71 payload containing PIN block or error code.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message 23 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Accept valid request and start the operation. Send the response when the operation completes. If successful, send Message 71. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message 24 PIN Entry Request With Custom Prompt

Format: <STX>24[Data]<ETX>[LRC] (request frame)
 <STX>71[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Starts secure online PIN entry using the secret key family.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>

24	2	Message ID
[Account]	8 to 19	PAN/account digits used for PIN block formatting.
<FS>	1	Field separator.
[Key/control]	Variable	Secret key id followed by optional PIN control and prompt data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
71	2	Message ID
[Status/PIN data]	Variable	Message 71 payload containing PIN block or error code.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message 24 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Accept valid request and start the operation. Send the response when the operation completes. If successful, send Message 71. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message 60 DUKPT PIN Entry Request

Format: <STX>60[Data]<ETX>[LRC] (request frame)
 <STX>71[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Starts online PIN entry using DUKPT.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
60	2	Message ID
[Account]	8 to 19	PAN/account digits.

<FS>	1	Field separator.
[Session key/control]	Variable	Session key, PIN length control, optional prompts, and optional timeout.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
71	2	Message ID
[PIN response]	Variable	Message 71 payload. Includes status, PIN block, KSN/key id, or error.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message 60 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Accept valid request and start the operation. Send the response when the operation completes. If successful, send Message 71. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message 70 PIN Entry Request

Format: <STX>70[Data]<ETX>[LRC] (request frame)
 <STX>71[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Starts DUKPT or MK/SK PIN entry. A leading dot selects MK/SK.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
70	2	Message ID
[Account]	8 to 19	PAN/account digits.
<FS>	1	Field separator.
[Session	Variable	Session key, PIN length control, optional prompts, and optional timeout.

key/control]	e	
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
71	2	Message ID
[PIN response]	Variable	Message 71 payload. Includes status, PIN block, KSN/key id, or error.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message 70 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Accept valid request and start the operation. Send the response when the operation completes. If successful, send Message 71. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message 7G PIN Change Request

Format: <STX>7G[Data]<ETX>[LRC] (request frame)
<STX>71[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Starts MK/SK PIN change style entry.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
7G	2	Message ID
[Account]	8 to 19	PAN/account digits.
<FS>	1	Field separator.
[Session key/control]	Variable	Session key, PIN length control, optional prompts, and optional timeout.
<ETX>	1	<03>

[LRC]	1	Checksum
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Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
71	2	Message ID
[PIN response]	Variable	Message 71 payload. Includes status, PIN block, KSN/key id, or error.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message 7G request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Accept valid request and start the operation. Send the response when the operation completes. If successful, send Message 71. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message Z60 External Prompt PIN Entry

Format: <STX>Z60[Data]<ETX>[LRC] (request frame)
<STX>71[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Starts PIN entry using an external prompt.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
Z60	3	Message ID
[Account]	8 to 19	PAN/account digits.
<FS>	1	Field separator.
[Session key/control]	Variable	Session key, PIN length control, optional prompts, and optional timeout.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
71	2	Message ID
[PIN response]	Variable	Message 71 payload. Includes status, PIN block, KSN/key id, or error.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message Z60 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Accept valid request and start the operation. Send the response when the operation completes. If successful, send Message 71. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message Z62 Custom Prompt PIN Entry

Format: <STX>Z62[Data]<ETX>[LRC] (request frame)

<STX>71[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Starts PIN entry with custom prompt lines and PIN limits.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
Z62	3	Message ID
[Account]	8 to 19	PAN/account digits.
<FS>	1	Field separator.
[Session key/control]	Variable	Session key, PIN length control, optional prompts, and optional timeout.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)

Field	Length	Value and description
<STX>	1	<02>
71	2	Message ID
[PIN response]	Variable	Message 71 payload. Includes status, PIN block, KSN/key id, or error.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message Z62 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Accept valid request and start the operation. Send the response when the operation completes. If successful, send Message 71. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message 62 Amount Authorization

Format: <STX>62[Data]<ETX>[LRC] (request frame)
 <STX>63[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Compatibility command that returns message 63.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
62	2	Message ID
[Data]	Variable	Compatibility payload.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
63	2	Message ID
[Status]	1	'0' accepted.

<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message 62 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message 63. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message 63 Response Message Compatibility Acknowledgement

Format: <STX>63[Data]<ETX>[LRC] (request frame)
<ACK>/<NAK>/<EOT> only unless an error frame is specified.

Message length: Variable.

Usage: Accepted as a no-op when a host sends a response-id command.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
63	2	Message ID
[Payload]	Variable	Ignored when received from host.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<ACK>	1	Command accepted
<NAK>	1	Transmission error
<EOT>	1	Format error, timeout, or canceled session

Message flow:

HOST	Dir.	PINPAD
Message 63 request frame	->	
	<-	<ACK>/<NAK>/<EOT>

HOST	Dir.	PINPAD
	<-	Apply command action. No data frame is sent for a normal accepted command.

Message 71 Response Message Compatibility Acknowledgement

Format: <STX>71[Data]<ETX>[LRC] (request frame)
 <ACK>/<NAK>/<EOT> only unless an error frame is specified.

Message length: Variable.

Usage: Accepted as a no-op when a host sends a response-id command.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
71	2	Message ID
[Payload]	Variable	Ignored when received from host.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<ACK>	1	Command accepted
<NAK>	1	Transmission error
<EOT>	1	Format error, timeout, or canceled session

Message flow:

HOST	Dir.	PINPAD
Message 71 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Apply command action. No data frame is sent for a normal accepted command.

Message 91 Response Message Compatibility Acknowledgement

Format: <STX>91[Data]<ETX>[LRC] (request frame)
 <ACK>/<NAK>/<EOT> only unless an error frame is specified.

Message length: Variable.

Usage: Accepted as a no-op when a host sends a response-id command.

Message element:

Request frame (HOST to PINPAD)		
--------------------------------	--	--

Field	Length	Value and description
<STX>	1	<02>
91	2	Message ID
[Payload]	Variable	Ignored when received from host.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<ACK>	1	Command accepted
<NAK>	1	Transmission error
<EOT>	1	Format error, timeout, or canceled session

Message flow:

HOST	Dir.	PINPAD
Message 91 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Apply command action. No data frame is sent for a normal accepted command.

Message 99 Response Message Compatibility Acknowledgement

Format: <STX>99[Data]<ETX>[LRC] (request frame)
 <ACK>/<NAK>/<EOT> only unless an error frame is specified.

Message length: Variable.

Usage: Accepted as a no-op when a host sends a response-id command.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
99	2	Message ID
[Payload]	Variable	Ignored when received from host.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description

<ACK>	1	Command accepted
<NAK>	1	Transmission error
<EOT>	1	Format error, timeout, or canceled session

Message flow:

HOST	Dir.	PINPAD
Message 99 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Apply command action. No data frame is sent for a normal accepted command.

Message Z65 Response Message Compatibility Acknowledgement

Format: <STX>Z65[Data]<ETX>[LRC] (request frame)
<ACK>/<NAK>/<EOT> only unless an error frame is specified.

Message length: Variable.

Usage: Accepted as a no-op when a host sends a response-id command.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
Z65	3	Message ID
[Payload]	Variable	Ignored when received from host.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<ACK>	1	Command accepted
<NAK>	1	Transmission error
<EOT>	1	Format error, timeout, or canceled session

Message flow:

HOST	Dir.	PINPAD
Message Z65 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Apply command action. No data frame is sent for a normal accepted command.

Message Z67 Response Message Compatibility Acknowledgement

Format: <STX>Z67[Data]<ETX>[LRC] (request frame)
 <ACK>/<NAK>/<EOT> only unless an error frame is specified.

Message length: Variable.

Usage: Accepted as a no-op when a host sends a response-id command.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
Z67	3	Message ID
[Payload]	Variable	Ignored when received from host.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<ACK>	1	Command accepted
<NAK>	1	Transmission error
<EOT>	1	Format error, timeout, or canceled session

Message flow:

HOST	Dir.	PINPAD
Message Z67 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Apply command action. No data frame is sent for a normal accepted command.

Message 7A Set DUKPT KSN Output Format

Format: <STX>7A[Data]<ETX>[LRC] (request frame)
 <ACK>/<NAK>/<EOT> only unless an error frame is specified.

Message length: Fixed.

Usage: Controls DUKPT KSN formatting.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
7A	2	Message ID
[Format]	1	KSN output format selector.

<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<ACK>	1	Command accepted
<NAK>	1	Transmission error
<EOT>	1	Format error, timeout, or canceled session

Message flow:

HOST	Dir.	PINPAD
Message 7A request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Apply command action. No data frame is sent for a normal accepted command.

Message 90 Load DUKPT Initial Key Set 0

Format: <STX>90[Data]<ETX>[LRC] (request frame)
 <STX>91[Data]<ETX>[LRC] (response frame)

Message length: Fixed.

Usage: Injects the first DUKPT key set.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
90	2	Message ID
[IPEK]	32	DUKPT initial key.
[KSN]	20	Initial key serial number.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
91	2	Message ID
[Status]	1	'0' success. '1' followed by reason on failure.
<ETX>	1	<03>

[LRC]	1	Checksum
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Message flow:

HOST	Dir.	PINPAD
Message 90 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message 91. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message 94 Load DUKPT Initial Key Set 1

Format: <STX>94[Data]<ETX>[LRC] (request frame)
<STX>91[Data]<ETX>[LRC] (response frame)

Message length: Fixed.

Usage: Injects the second DUKPT key set.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
94	2	Message ID
[IPEK]	32	DUKPT initial key.
[KSN]	20	Initial key serial number.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
91	2	Message ID
[Status]	1	'0' success. '1' followed by reason on failure.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message 94 request frame	->	

HOST	Dir.	PINPAD
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message 91. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message 96 Select DUKPT Key Set

Format: <STX>96[Data]<ETX>[LRC] (request frame)
 <ACK>/<NAK>/<EOT> only unless an error frame is specified.

Message length: Fixed.

Usage: Selects active DUKPT key set.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
96	2	Message ID
[Key set]	1	'0' or '1'.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<ACK>	1	Command accepted
<NAK>	1	Transmission error
<EOT>	1	Format error, timeout, or canceled session

Message flow:

HOST	Dir.	PINPAD
Message 96 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Apply command action. No data frame is sent for a normal accepted command.

Message 98 Query DUKPT Key Set

Format: <STX>98[Data]<ETX>[LRC] (request frame)
 <STX>99[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Queries DUKPT key state.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
98	2	Message ID
[Key set]	1	'0' or '1'.
[Info flag]	0 or 1	Optional '1' requests extra key information.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
99	2	Message ID
[Status]	1	Key state.
[Info]	Variable	Optional algorithm/type/KCV style fields separated by <FS>.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message 98 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message 99. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message Z64 Query Master Key KCV

Format: <STX>Z64[Data]<ETX>[LRC] (request frame)
<STX>Z65[Data]<ETX>[LRC] (response frame)

Message length: Fixed.

Usage: Calculates the master key check value.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
Z64	3	Message ID
[Key ID]	1	Master key id.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
Z65	3	Message ID
[Key ID]	1	Echoed key id.
[KCV]	6	Three-byte key check value, or '?' when unavailable.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message Z64 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message Z65. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message Z66 Generate MAC

Format: <STX>Z66[Data]<ETX>[LRC] (request frame)
 <STX>Z67[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Generates a MAC using the active MAC-capable key.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
Z66	3	Message ID
[MAC data]	Variable	Data to MAC according to active key metadata.

	e	
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
Z67	3	Message ID
[Status]	1	'0' success, '2' failure.
[MAC]	16	MAC value when successful.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message Z66 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message Z67. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Section 4 Magnetic Stripe and Multi-Interface Messages

Message Q1 Arm Magnetic Stripe Reader

Format: <STX>Q1[Data]<ETX>[LRC] (request frame)
 <STX>81[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Displays the swipe prompt and waits for a swipe. Cached multi-interface swipe data is returned immediately when available.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
Q1	2	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
81	2	Message ID
[Track 1]	Variable	Track 1 data when present.
<FS>	1	Field separator.
[Track 2]	Variable	Track 2 data when present.
[Track 3]	Variable	Track 3 data when present.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message Q1 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Accept valid request and start the operation. Send the response when the operation completes. If successful, send Message 81. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message Q8 Read Contactless Track Data

Format: <STX>Q8[Data]<ETX>[LRC] (request frame)
 <STX>83[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Starts RF card read and returns track-equivalent data.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
Q8	2	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
83	2	Message ID
[Track 1]	Variable	Track 1 data when present.
<FS>	1	Field separator.
[Track 2]	Variable	Track 2 data when present.
[Track 3]	Variable	Track 3 data when present.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message Q8 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Accept valid request and start the operation. Send the response when the operation completes. If successful, send Message 83. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message Q9 Read Swipe or Contactless Track Data

Format: <STX>Q9[Data]<ETX>[LRC] (request frame)

<STX>81/83[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Arms both swipe and RF track paths. The first successful reader wins.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
Q9	2	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
81/83	5	Message ID
[Track 1]	Variable	Track 1 data when present.
<FS>	1	Field separator.
[Track 2]	Variable	Track 2 data when present.
[Track 3]	Variable	Track 3 data when present.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message Q9 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Accept valid request and start the operation. Send the response when the operation completes. If successful, send Message 81/83. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message QF Arm Magnetic Stripe Reader With Image Prompt

Format: <STX>QF[Data]<ETX>[LRC] (request frame)

<STX>81[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Compatibility image-prompt variant. Android prompt animation is used.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
QF	2	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
81	2	Message ID
[Track 1]	Variable	Track 1 data when present.
<FS>	1	Field separator.
[Track 2]	Variable	Track 2 data when present.
[Track 3]	Variable	Track 3 data when present.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message QF request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Accept valid request and start the operation. Send the response when the operation completes. If successful, send Message 81. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message QG Read Contactless Track Data With Image Prompt

Format: <STX>QG[Data]<ETX>[LRC] (request frame)

<STX>83[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Compatibility image-prompt variant. Android prompt animation is used.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
QG	2	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
83	2	Message ID
[Track 1]	Variable	Track 1 data when present.
<FS>	1	Field separator.
[Track 2]	Variable	Track 2 data when present.
[Track 3]	Variable	Track 3 data when present.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message QG request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Accept valid request and start the operation. Send the response when the operation completes. If successful, send Message 83. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message QH Read Swipe or Contactless Track Data With Image Prompt

Format: <STX>QH[Data]<ETX>[LRC] (request frame)
 <STX>81/83[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Compatibility dual-reader image-prompt variant.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
QH	2	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
81/83	5	Message ID
[Track 1]	Variable	Track 1 data when present.
<FS>	1	Field separator.
[Track 2]	Variable	Track 2 data when present.
[Track 3]	Variable	Track 3 data when present.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message QH request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Accept valid request and start the operation. Send the response when the operation completes. If successful, send Message 81/83. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message Q2 Transaction Complete

Format: <STX>Q2[Data]<ETX>[LRC] (request frame)
<ACK>/<NAK>/<EOT> only unless an error frame is specified.

Message length: Variable.

Usage: Completes a magnetic stripe/card-read session and returns to idle.

Message element:

Request frame (HOST to PINPAD)

Field	Length	Value and description
<STX>	1	<02>
Q2	2	Message ID
[Option]	Variable	Command-specific option.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<ACK>	1	Command accepted
<NAK>	1	Transmission error
<EOT>	1	Format error, timeout, or canceled session

Message flow:

HOST	Dir.	PINPAD
Message Q2 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Apply command action. No data frame is sent for a normal accepted command.

Message Q3 Ignore Card Swipe

Format: <STX>Q3[Data]<ETX>[LRC] (request frame)
 <ACK>/<NAK>/<EOT> only unless an error frame is specified.

Message length: Variable.

Usage: Stops active swipe handling and ignores the current swipe.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
Q3	2	Message ID
[Option]	Variable	Command-specific option.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description

<ACK>	1	Command accepted
<NAK>	1	Transmission error
<EOT>	1	Format error, timeout, or canceled session

Message flow:

HOST	Dir.	PINPAD
Message Q3 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Apply command action. No data frame is sent for a normal accepted command.

Message Q4 Set Swipe Track Mode

Format: <STX>Q4[Data]<ETX>[LRC] (request frame)
<ACK>/<NAK>/<EOT> only unless an error frame is specified.

Message length: Variable.

Usage: Sets swipe track mode: 0 track 2 only, 1 disabled, 2 all tracks.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
Q4	2	Message ID
[Option]	Variable	Command-specific option.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<ACK>	1	Command accepted
<NAK>	1	Transmission error
<EOT>	1	Format error, timeout, or canceled session

Message flow:

HOST	Dir.	PINPAD
Message Q4 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Apply command action. No data frame is sent for a normal accepted command.

Message Q5 Set Swipe Retry Count

Format: <STX>Q5[Data]<ETX>[LRC] (request frame)
 <ACK>/<NAK>/<EOT> only unless an error frame is specified.

Message length: Variable.

Usage: Sets retry count 0 through 9.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
Q5	2	Message ID
[Option]	Variable	Command-specific option.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<ACK>	1	Command accepted
<NAK>	1	Transmission error
<EOT>	1	Format error, timeout, or canceled session

Message flow:

HOST	Dir.	PINPAD
Message Q5 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Apply command action. No data frame is sent for a normal accepted command.

Message QA Set Contactless Track Mode

Format: <STX>QA[Data]<ETX>[LRC] (request frame)
 <ACK>/<NAK>/<EOT> only unless an error frame is specified.

Message length: Variable.

Usage: Sets contactless track mode: 0 track 2 only, 1 disabled, 2 all tracks.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
QA	2	Message ID
[Option]	Variable	Command-specific option.

	e	
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<ACK>	1	Command accepted
<NAK>	1	Transmission error
<EOT>	1	Format error, timeout, or canceled session

Message flow:

HOST	Dir.	PINPAD
Message QA request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Apply command action. No data frame is sent for a normal accepted command.

Message Q6 Set Swipe Output Format

Format: <STX>Q6[Data]<ETX>[LRC] (request frame)
 <STX>Q6[Data]<ETX>[LRC] (response frame)

Message length: Fixed.

Usage: Sets magnetic stripe output format.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
Q6	2	Message ID
[Format]	1	'0' clear text, '2' clear text with sentinels.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
Q6	2	Message ID
Q6	2	Request frame echoed on success.
<ETX>	1	<03>

[LRC]	1	Checksum
-------	---	----------

Message flow:

HOST	Dir.	PINPAD
Message Q6 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message Q6. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message Q7 Query Swipe Output Format

Format: <STX>Q7[Data]<ETX>[LRC] (request frame)
 <STX>Q7[Data]<ETX>[LRC] (response frame)

Message length: Fixed.

Usage: Returns magnetic stripe output format.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
Q7	2	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
Q7	2	Message ID
[Format]	1	Current output format.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message Q7 request frame	->	
	<-	<ACK>/<NAK>/<EOT>

HOST	Dir.	PINPAD
	<-	Validate command format and process request. If successful, send Message Q7. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message QB Set Swipe Auto Arm

Format: <STX>QB[Data]<ETX>[LRC] (request frame)
 <STX>QB[Data]<ETX>[LRC] (response frame)

Message length: Fixed.

Usage: Enables or disables automatic swipe arming.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
QB	2	Message ID
[Option]	1	'0' disable, '1' enable.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
QB	2	Message ID
[Status]	1	'0' success, '1' failure.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message QB request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message QB. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>.	->	

HOST	Dir.	PINPAD
If canceling the session, send <EOT>.		

Message QC Set Contactless Auto Arm

Format: <STX>QC[Data]<ETX>[LRC] (request frame)
 <STX>QC[Data]<ETX>[LRC] (response frame)

Message length: Fixed.

Usage: Enables or disables contactless auto-arm compatibility flag.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
QC	2	Message ID
[Option]	1	'0' disable, '1' enable.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
QC	2	Message ID
[Status]	1	'0' success, '1' failure.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message QC request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message QC. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message QD Set Contactless Output Format

Format: <STX>QD[Data]<ETX>[LRC] (request frame)
 <STX>QD[Data]<ETX>[LRC] (response frame)

Message length: Fixed.

Usage: Sets contactless track-equivalent output format.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
QD	2	Message ID
[Format]	1	'0' clear text, '2' clear text with sentinels.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
QD	2	Message ID
QD	2	Request frame echoed on success.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message QD request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message QD. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message QI Query Contactless Tags

Format: <STX>QI[Data]<ETX>[LRC] (request frame)
<STX>QJ[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Returns requested contactless transaction data objects.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>

QI	2	Message ID
[DOL]	Variable	Tag list or data object list. Optional ready-control byte may precede it.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
QJ	2	Message ID
[Status/control]	Variable	'F' for no data or '11' plus TLV data.
<FS>	1	Field separator before TLV data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message QI request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message QJ. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message QK Multi-Interface Card Detection

Format: <STX>QK[Data]<ETX>[LRC] (request frame)
<STX>QK[Data]<ETX>[LRC] (response frame)

Message length: Fixed.

Usage: Prompts for swipe, insert, or tap and returns the detected interface.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
QK	2	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
QK	2	Message ID
[Result]	1	'1' swipe, '2' chip, '3' contactless, '0' error/cancel.
[Error]	0 or 1	Error code when result is 0.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message QK request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Accept valid request and start the operation. Send the response when the operation completes. If successful, send Message QK. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message 81 Magnetic Stripe Data

Format: <STX>81[Data]<ETX>[LRC] (request frame)
 <STX>81[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Device-to-host track-data response.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
81	2	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
81	2	Message ID

[Track data]	Variable	Track 1, track 2, and track 3 separated by <FS>.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message 81 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message 81. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message 83 Contactless Track Data

Format: <STX>83[Data]<ETX>[LRC] (request frame)
 <STX>83[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Device-to-host RF track-equivalent response.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
83	2	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
83	2	Message ID
[Track data]	Variable	Track-equivalent data separated by <FS>.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
------	------	--------

HOST	Dir.	PINPAD
Message 83 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message 83. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Section 5 CPU Smart Card Messages

Message I00 Query Primary Smart Card Presence

Format: <STX>I00[Data]<ETX>[LRC] (request frame)
 <STX>I00[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Checks ICC card presence.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
I00	3	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
I00	3	Message ID
[Status]	1	'F' present, '0' absent.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message I00 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message I00. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message I01 Primary Smart Card Cold Reset

Format: <STX>I01[Data]<ETX>[LRC] (request frame)
 <STX>I02[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Powers on ICC card and returns ATR.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
I01	3	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
I02	3	Message ID
[ATR]	Variable	ATR returned in message I02.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message I01 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message I02. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message I04 Primary Smart Card Deactivate

Format: <STX>I04[Data]<ETX>[LRC] (request frame)
 <STX>I04[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Powers off ICC card.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
I04	3	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
I04	3	Message ID
[Status]	0	No data on success.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message I04 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message I04. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message I06 Primary Smart Card APDU Exchange

Format: <STX>I06[Data]<ETX>[LRC] (request frame)
 <STX>I07[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Exchanges an APDU with the powered ICC card.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
I06	3	Message ID
[APDU]	8 to 524	APDU command in hexadecimal.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>

I07	3	Message ID
[APDU response]	Variable	Response APDU returned in message I07.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message I06 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message I07. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message I11 SAM Cold Reset

Format: <STX>I11[Data]<ETX>[LRC] (request frame)
<STX>I12[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Powers on selected SAM and returns ATR.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
I11	3	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
I12	3	Message ID
[ATR]	Variable	ATR returned in message I12.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message I11 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message I12. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message I14 SAM Deactivate

Format: <STX>I14[Data]<ETX>[LRC] (request frame)
<STX>I14[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Powers off selected SAM.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
I14	3	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
I14	3	Message ID
[Status]	0	No data on success.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message I14 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message I14. If format is invalid, send <EOT> or command-specific failure.
Verify response frame.	->	

HOST	Dir.	PINPAD
If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.		

Message I15 Select SAM Interface

Format: <STX>I15[Data]<ETX>[LRC] (request frame)
<STX>I15[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Selects active SAM slot.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
I15	3	Message ID
[Interface]	1	'1' PSAM1, '2' PSAM2.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
I15	3	Message ID
[Interface]	1	Selected interface echoed.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message I15 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message I15. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message I16 SAM APDU Exchange

Format: <STX>I16[Data]<ETX>[LRC] (request frame)

<STX>I17[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Exchanges an APDU with the powered SAM.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
I16	3	Message ID
[APDU]	8 to 524	APDU command in hexadecimal.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
I17	3	Message ID
[APDU response]	Variable	Response APDU returned in message I17.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message I16 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message I17. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message I02 CPU Card Compatibility or Response Identifier

Format: <STX>I02[Data]<ETX>[LRC] (request frame)

<STX>I0F[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Reserved, unsupported, or device response identifier.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
I02	3	Message ID
[Payload]	Variable	Ignored or unsupported when received from host.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
I0F	3	Message ID
[Error]	2	I0F error code for unsupported host request.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message I02 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message I0F. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message I03 CPU Card Compatibility or Response Identifier

Format: <STX>I03[Data]<ETX>[LRC] (request frame)
 <STX>I0F[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Reserved, unsupported, or device response identifier.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
I03	3	Message ID
[Payload]	Variable	Ignored or unsupported when received from host.

	e	
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
I0F	3	Message ID
[Error]	2	I0F error code for unsupported host request.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message I03 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message I0F. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message I05 CPU Card Compatibility or Response Identifier

Format: <STX>I05[Data]<ETX>[LRC] (request frame)
<STX>I0F[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Reserved, unsupported, or device response identifier.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
I05	3	Message ID
[Payload]	Variable	Ignored or unsupported when received from host.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
I0F	3	Message ID
[Error]	2	I0F error code for unsupported host request.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message I05 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message I0F. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message I07 CPU Card Compatibility or Response Identifier

Format: <STX>I07[Data]<ETX>[LRC] (request frame)
 <STX>I0F[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Reserved, unsupported, or device response identifier.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
I07	3	Message ID
[Payload]	Variable	Ignored or unsupported when received from host.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
I0F	3	Message ID
[Error]	2	I0F error code for unsupported host request.

<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message I07 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message I0F. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message I08 CPU Card Compatibility or Response Identifier

Format: <STX>I08[Data]<ETX>[LRC] (request frame)
<STX>I0F[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Reserved, unsupported, or device response identifier.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
I08	3	Message ID
[Payload]	Variable	Ignored or unsupported when received from host.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
I0F	3	Message ID
[Error]	2	I0F error code for unsupported host request.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
------	------	--------

HOST	Dir.	PINPAD
Message I08 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message I0F. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message I09 CPU Card Compatibility or Response Identifier

Format: <STX>I09[Data]<ETX>[LRC] (request frame)
<STX>I0F[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Reserved, unsupported, or device response identifier.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
I09	3	Message ID
[Payload]	Variable	Ignored or unsupported when received from host.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
I0F	3	Message ID
[Error]	2	I0F error code for unsupported host request.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message I09 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message I0F. If format is invalid, send <EOT> or

HOST	Dir.	PINPAD
		command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message I0A CPU Card Compatibility or Response Identifier

Format: <STX>I0A[Data]<ETX>[LRC] (request frame)
 <STX>I0F[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Reserved, unsupported, or device response identifier.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
I0A	3	Message ID
[Payload]	Variable	Ignored or unsupported when received from host.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
I0F	3	Message ID
[Error]	2	I0F error code for unsupported host request.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message I0A request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message I0F. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message I0B CPU Card Compatibility or Response Identifier

Format: <STX>I0B[Data]<ETX>[LRC] (request frame)
 <STX>I0F[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Reserved, unsupported, or device response identifier.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
I0B	3	Message ID
[Payload]	Variable	Ignored or unsupported when received from host.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
I0F	3	Message ID
[Error]	2	I0F error code for unsupported host request.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message I0B request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message I0F. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message I0C CPU Card Compatibility or Response Identifier

Format: <STX>I0C[Data]<ETX>[LRC] (request frame)
 <STX>I0F[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Reserved, unsupported, or device response identifier.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
I0C	3	Message ID
[Payload]	Variable	Ignored or unsupported when received from host.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
I0F	3	Message ID
[Error]	2	I0F error code for unsupported host request.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message I0C request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message I0F. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message I0D CPU Card Compatibility or Response Identifier

Format: <STX>I0D[Data]<ETX>[LRC] (request frame)
<STX>I0F[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Reserved, unsupported, or device response identifier.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>

I0D	3	Message ID
[Payload]	Variable	Ignored or unsupported when received from host.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
I0F	3	Message ID
[Error]	2	I0F error code for unsupported host request.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message I0D request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message I0F. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message I0E CPU Card Compatibility or Response Identifier

Format: <STX>I0E[Data]<ETX>[LRC] (request frame)
<STX>I0F[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Reserved, unsupported, or device response identifier.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
I0E	3	Message ID
[Payload]	Variable	Ignored or unsupported when received from host.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
I0F	3	Message ID
[Error]	2	I0F error code for unsupported host request.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message I0E request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message I0F. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message I0F CPU Card Compatibility or Response Identifier

Format: <STX>I0F[Data]<ETX>[LRC] (request frame)
<STX>I0F[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Reserved, unsupported, or device response identifier.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
I0F	3	Message ID
[Payload]	Variable	Ignored or unsupported when received from host.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
I0F	3	Message ID

[Error]	2	I0F error code for unsupported host request.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message I0F request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message I0F. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message I12 CPU Card Compatibility or Response Identifier

Format: <STX>I12[Data]<ETX>[LRC] (request frame)
<STX>I0F[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Reserved, unsupported, or device response identifier.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
I12	3	Message ID
[Payload]	Variable	Ignored or unsupported when received from host.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
I0F	3	Message ID
[Error]	2	I0F error code for unsupported host request.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message I12 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message I0F. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message I17 CPU Card Compatibility or Response Identifier

Format: <STX>I17[Data]<ETX>[LRC] (request frame)
<STX>I0F[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Reserved, unsupported, or device response identifier.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
I17	3	Message ID
[Payload]	Variable	Ignored or unsupported when received from host.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
I0F	3	Message ID
[Error]	2	I0F error code for unsupported host request.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message I17 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message I0F. If format is invalid, send <EOT> or

HOST	Dir.	PINPAD
		command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Section 6 RF Memory Card Messages

Message P01 Enable or Disable RF Memory Card Reader

Format: <STX>P01[Data]<ETX>[LRC] (request frame)
 <STX>P01[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Opens or closes the RF reader.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
P01	3	Message ID
[Data]	Variable	[0/1]
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
P01	3	Message ID
[Status]	1	'0' success, '1' failure.
[Data]	Variable	Command-specific response data when successful.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message P01 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message P01. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message P02 Query RF Card Presence

Format: <STX>P02[Data]<ETX>[LRC] (request frame)
 <STX>P02[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Returns ATQA when a Type A card is present.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
P02	3	Message ID
[Data]	Variable	No payload.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
P02	3	Message ID
[Status]	1	'0' success, '1' failure.
[Data]	Variable	Command-specific response data when successful.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message P02 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message P02. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message P03 RF Anticollision

Format: <STX>P03[Data]<ETX>[LRC] (request frame)
 <STX>P03[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Returns UID.

Message element:

Request frame (HOST to PINPAD)

Field	Length	Value and description
<STX>	1	<02>
P03	3	Message ID
[Data]	Variable	No payload.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
P03	3	Message ID
[Status]	1	'0' success, '1' failure.
[Data]	Variable	Command-specific response data when successful.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message P03 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message P03. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message P04 RF Select

Format: <STX>P04[Data]<ETX>[LRC] (request frame)
 <STX>P04[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Returns SAK.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
P04	3	Message ID

[Data]	Variable	No payload.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
P04	3	Message ID
[Status]	1	'0' success, '1' failure.
[Data]	Variable	Command-specific response data when successful.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message P04 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message P04. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message P05 RF Activate

Format: <STX>P05[Data]<ETX>[LRC] (request frame)
 <STX>P05[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Returns ATQA, SAK, and UID.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
P05	3	Message ID
[Data]	Variable	No payload.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
P05	3	Message ID
[Status]	1	'0' success, '1' failure.
[Data]	Variable	Command-specific response data when successful.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message P05 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message P05. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message P06 RF Halt

Format: <STX>P06[Data]<ETX>[LRC] (request frame)
<STX>P06[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Halts or deselects the current card.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
P06	3	Message ID
[Data]	Variable	No payload.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description

<STX>	1	<02>
P06	3	Message ID
[Status]	1	'0' success, '1' failure.
[Data]	Variable	Command-specific response data when successful.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message P06 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message P06. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message P07 Authenticate Classic Sector

Format: <STX>P07[Data]<ETX>[LRC] (request frame)
 <STX>P07[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Authenticates a Classic sector using direct or stored key.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
P07	3	Message ID
[Data]	Variable	[Sector][Key data]
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
P07	3	Message ID
[Status]	1	'0' success, '1' failure.

[Data]	Variable	Command-specific response data when successful.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message P07 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message P07. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message P08 Read Ultralight Page

Format: <STX>P08[Data]<ETX>[LRC] (request frame)
 <STX>P08[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Reads a 4-byte Ultralight page.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
P08	3	Message ID
[Data]	Variable	[Page]
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
P08	3	Message ID
[Status]	1	'0' success, '1' failure.
[Data]	Variable	Command-specific response data when successful.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message P08 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message P08. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message P09 Write Ultralight Page

Format: <STX>P09[Data]<ETX>[LRC] (request frame)
<STX>P09[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Writes a 4-byte Ultralight page.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
P09	3	Message ID
[Data]	Variable	[Page][Data]
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
P09	3	Message ID
[Status]	1	'0' success, '1' failure.
[Data]	Variable	Command-specific response data when successful.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
------	------	--------

HOST	Dir.	PINPAD
Message P09 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message P09. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message P10 Read Block

Format: <STX>P10[Data]<ETX>[LRC] (request frame)
 <STX>P10[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Reads a block or page.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
P10	3	Message ID
[Data]	Variable	[Block]
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
P10	3	Message ID
[Status]	1	'0' success, '1' failure.
[Data]	Variable	Command-specific response data when successful.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message P10 request frame	->	
	<-	<ACK>/<NAK>/<EOT>

HOST	Dir.	PINPAD
	<-	Validate command format and process request. If successful, send Message P10. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message P11 Write Block

Format: <STX>P11[Data]<ETX>[LRC] (request frame)
 <STX>P11[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Writes a block or page.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
P11	3	Message ID
[Data]	Variable	[Block][Data]
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
P11	3	Message ID
[Status]	1	'0' success, '1' failure.
[Data]	Variable	Command-specific response data when successful.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message P11 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message P11. If format is invalid, send <EOT> or command-specific failure.

HOST	Dir.	PINPAD
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message P12 Read Sector

Format: <STX>P12[Data]<ETX>[LRC] (request frame)
 <STX>P12[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Reads all blocks in a sector.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
P12	3	Message ID
[Data]	Variable	[Sector]
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
P12	3	Message ID
[Status]	1	'0' success, '1' failure.
[Data]	Variable	Command-specific response data when successful.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message P12 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message P12. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>.	->	

HOST	Dir.	PINPAD
If canceling the session, send <EOT>.		

Message P13 Write Sector

Format: <STX>P13[Data]<ETX>[LRC] (request frame)
 <STX>P13[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Writes sequential blocks in a sector.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
P13	3	Message ID
[Data]	Variable	[Sector][Data]
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
P13	3	Message ID
[Status]	1	'0' success, '1' failure.
[Data]	Variable	Command-specific response data when successful.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message P13 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message P13. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message P14 Value Operation

Format: <STX>P14[Data]<ETX>[LRC] (request frame)
 <STX>P14[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Performs value write, increment, decrement, or backup operation.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
P14	3	Message ID
[Data]	Variable	[Block][Mode][Value/Target]
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
P14	3	Message ID
[Status]	1	'0' success, '1' failure.
[Data]	Variable	Command-specific response data when successful.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message P14 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message P14. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message P15 Load RF Key

Format: <STX>P15[Data]<ETX>[LRC] (request frame)
 <STX>P15[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Stores key A and key B for later authentication.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
P15	3	Message ID
[Data]	Variable	[KeyNo][KeyA][KeyB]
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
P15	3	Message ID
[Status]	1	'0' success, '1' failure.
[Data]	Variable	Command-specific response data when successful.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message P15 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message P15. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message P16 Identify RF Card Type

Format: <STX>P16[Data]<ETX>[LRC] (request frame)
<STX>P16[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Returns card type code.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description

<STX>	1	<02>
P16	3	Message ID
[Data]	Variable	No payload.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
P16	3	Message ID
[Status]	1	'0' success, '1' failure.
[Data]	Variable	Command-specific response data when successful.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message P16 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message P16. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message P17 Activate DESFire

Format: <STX>P17[Data]<ETX>[LRC] (request frame)
 <STX>P17[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Activates DESFire and returns ATS or UID.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
P17	3	Message ID
[Data]	Variable	No payload.

<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
P17	3	Message ID
[Status]	1	'0' success, '1' failure.
[Data]	Variable	Command-specific response data when successful.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message P17 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message P17. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message P18 Deselect DESFire

Format: <STX>P18[Data]<ETX>[LRC] (request frame)
<STX>P18[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Deselects DESFire and clears RF state.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
P18	3	Message ID
[Data]	Variable	No payload.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
P18	3	Message ID
[Status]	1	'0' success, '1' failure.
[Data]	Variable	Command-specific response data when successful.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message P18 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message P18. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message P19 RF APDU Exchange

Format: <STX>P19[Data]<ETX>[LRC] (request frame)
 <STX>P19[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Exchanges APDU or raw command depending on detected RF type.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
P19	3	Message ID
[Data]	Variable	[APDU]
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>

P19	3	Message ID
[Status]	1	'0' success, '1' failure.
[Data]	Variable	Command-specific response data when successful.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message P19 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message P19. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message P20 RF Raw Block Exchange

Format: <STX>P20[Data]<ETX>[LRC] (request frame)
<STX>P20[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Performs raw RF command exchange.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
P20	3	Message ID
[Data]	Variable	[CRC][Wait][Data]
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
P20	3	Message ID
[Status]	1	'0' success, '1' failure.
[Data]	Variable	Command-specific response data when successful.

<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message P20 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message P20. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Section 7 Display, Prompt, Data Entry, and Image Messages

Message B1 Set Display Font Size

Format: <STX>B1[Data]<ETX>[LRC] (request frame)
 <STX>B2[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Sets display font size compatibility option.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
B1	2	Message ID
[Data]	Variable	[Option]
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
B2	2	Message ID
[Status/Data]	Variable	Command-specific status or data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message B1 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message B2. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message B3 Set Display Font Color

Format: <STX>B3[Data]<ETX>[LRC] (request frame)
 <STX>B4[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Sets display font color compatibility option.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
B3	2	Message ID
[Data]	Variable	[Foreground][Background]
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
B4	2	Message ID
[Status/Data]	Variable	Command-specific status or data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message B3 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message B4. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message BB Set Screen Saver

Format: <STX>BB[Data]<ETX>[LRC] (request frame)
<STX>BC[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Sets screen saver compatibility option.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>

BB	2	Message ID
[Data]	Variable	[WaitTime][Type]
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
BC	2	Message ID
[Status/Data]	Variable	Command-specific status or data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message BB request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message BC. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message BD Enable or Disable Screen Saver

Format: <STX>BD[Data]<ETX>[LRC] (request frame)
 <STX>BE[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Enables or disables screen saver compatibility option.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
BD	2	Message ID
[Data]	Variable	[0/1]
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
BE	2	Message ID
[Status/Data]	Variable	Command-specific status or data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message BD request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message BE. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message J0 Initialize Image Table

Format: <STX>J0[Data]<ETX>[LRC] (request frame)
 <STX>J0[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Initializes local image table.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
J0	2	Message ID
[Data]	Variable	No payload.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>

J0	2	Message ID
[Status/Data]	Variable	Command-specific status or data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message J0 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message J0. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message J1 Query Image Table

Format: <STX>J1[Data]<ETX>[LRC] (request frame)
<STX>J1[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Returns image table entries.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
J1	2	Message ID
[Data]	Variable	No payload.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
J1	2	Message ID
[Status/Data]	Variable	Command-specific status or data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message J1 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message J1. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message J2 Select Images

Format: <STX>J2[Data]<ETX>[LRC] (request frame)
<STX>J2[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Selects stored image names.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
J2	2	Message ID
[Data]	Variable	[Control][Names]
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
J2	2	Message ID
[Status/Data]	Variable	Command-specific status or data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message J2 request frame	->	

HOST	Dir.	PINPAD
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message J2. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message J3 Delete Images

Format: <STX>J3[Data]<ETX>[LRC] (request frame)
 <STX>J3[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Deletes stored images.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
J3	2	Message ID
[Data]	Variable	[Names]
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
J3	2	Message ID
[Status/Data]	Variable	Command-specific status or data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message J3 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message J3. If format is invalid, send <EOT> or command-specific failure.

HOST	Dir.	PINPAD
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message J4 Download Image Packet

Format: <STX>J4[Data]<ETX>[LRC] (request frame)
<STX>J4[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Stores image packet data.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
J4	2	Message ID
[Data]	Variable	[Packet]
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
J4	2	Message ID
[Status/Data]	Variable	Command-specific status or data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message J4 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message J4. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message J5 Upload Image Packet

Format: <STX>J5[Data]<ETX>[LRC] (request frame)
 <STX>J5[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Returns stored image packet data.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
J5	2	Message ID
[Data]	Variable	[Control][Name]
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
J5	2	Message ID
[Status/Data]	Variable	Command-specific status or data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message J5 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message J5. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message J7 Set Idle Image

Format: <STX>J7[Data]<ETX>[LRC] (request frame)
 <STX>J7[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Accepted for compatibility.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
J7	2	Message ID
[Data]	Variable	[Name]
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
J7	2	Message ID
[Status/Data]	Variable	Command-specific status or data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message J7 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message J7. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message J8 Enable Idle Image

Format: <STX>J8[Data]<ETX>[LRC] (request frame)
 <STX>J8[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Accepted for compatibility.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>

J8	2	Message ID
[Data]	Variable	[Option]
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
J8	2	Message ID
[Status/Data]	Variable	Command-specific status or data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message J8 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message J8. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message J9 Show Image

Format: <STX>J9[Data]<ETX>[LRC] (request frame)
 <STX>J9[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Shows stored image if available.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
J9	2	Message ID
[Data]	Variable	[Name]
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
J9	2	Message ID
[Status/Data]	Variable	Command-specific status or data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message J9 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message J9. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message JA Download Boot Logo

Format: <STX>JA[Data]<ETX>[LRC] (request frame)
 <STX>JA[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Stores boot logo packet data.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
JA	2	Message ID
[Data]	Variable	[Packet]
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>

JA	2	Message ID
[Status/Data]	Variable	Command-specific status or data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message JA request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message JA. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message BF Preview Screen Saver

Format: <STX>BF[Data]<ETX>[LRC] (request frame)
<ACK>/<NAK>/<EOT> only unless an error frame is specified.

Message length: Variable.

Usage: Compatibility no-op.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
BF	2	Message ID
[Data]	Variable	Command-specific payload.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<ACK>	1	Command accepted
<NAK>	1	Transmission error
<EOT>	1	Format error, timeout, or canceled session

Message flow:

HOST	Dir.	PINPAD
------	------	--------

HOST	Dir.	PINPAD
Message BF request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Apply command action. No data frame is sent for a normal accepted command.

Message J6 Play Selected Images

Format: <STX>J6[Data]<ETX>[LRC] (request frame)
 <ACK>/<NAK>/<EOT> only unless an error frame is specified.

Message length: Variable.

Usage: Displays selected image sequence if available.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
J6	2	Message ID
[Data]	Variable	Command-specific payload.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<ACK>	1	Command accepted
<NAK>	1	Transmission error
<EOT>	1	Format error, timeout, or canceled session

Message flow:

HOST	Dir.	PINPAD
Message J6 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Apply command action. No data frame is sent for a normal accepted command.

Message Z0 Move Display Cursor

Format: <STX>Z0[Data]<ETX>[LRC] (request frame)
 <STX>Z0[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Compatibility cursor command.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
Z0	2	Message ID
[Data]	Variable	[Line]
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
Z0	2	Message ID
[Status/Data]	Variable	Command-specific status, key code, text, or no frame depending on command.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message Z0 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message Z0. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message Z1 Reset Display and Transaction State

Format: <STX>Z1[Data]<ETX>[LRC] (request frame)
 <STX>Z1[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Clears prompts, transaction amount display, cached card data, active reads, and active PIN entry.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>

Z1	2	Message ID
[Data]	Variable	No payload.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
Z1	2	Message ID
[Status/Data]	Variable	Command-specific status, key code, text, or no frame depending on command.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message Z1 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message Z1. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message 72 Cancel Transaction

Format: <STX>72[Data]<ETX>[LRC] (request frame)
 <STX>72[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Cancels active card or PIN operation and returns to idle.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
72	2	Message ID
[Data]	Variable	No payload.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
72	2	Message ID
[Status/Data]	Variable	Command-specific status, key code, text, or no frame depending on command.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message 72 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message 72. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message Z2 Display Prompt

Format: <STX>Z2[Data]<ETX>[LRC] (request frame)
 <STX>Z2[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Displays one fixed or custom prompt.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
Z2	2	Message ID
[Data]	Variable	[Prompt]
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>

Z2	2	Message ID
[Status/Data]	Variable	Command-specific status, key code, text, or no frame depending on command.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message Z2 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message Z2. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message Z3 Display Multi-Line Prompt

Format: <STX>Z3[Data]<ETX>[LRC] (request frame)
 <STX>Z3[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Displays fixed or custom prompt lines.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
Z3	2	Message ID
[Data]	Variable	[Prompts]
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
Z3	2	Message ID
[Status/Data]	Variable	Command-specific status, key code, text, or no frame depending on command.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message Z3 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message Z3. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message Z7 Cancel Message Display Option

Format: <STX>Z7[Data]<ETX>[LRC] (request frame)
<STX>Z7[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Controls whether cancel message is shown.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
Z7	2	Message ID
[Data]	Variable	[0/1]
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
Z7	2	Message ID
[Status/Data]	Variable	Command-specific status, key code, text, or no frame depending on command.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message Z7 request frame	->	

HOST	Dir.	PINPAD
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message Z7. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message Z8 Set Idle Prompt

Format: <STX>Z8[Data]<ETX>[LRC] (request frame)
 <STX>Z8[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Sets the idle prompt text.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
Z8	2	Message ID
[Data]	Variable	[Prompt]
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
Z8	2	Message ID
[Status/Data]	Variable	Command-specific status, key code, text, or no frame depending on command.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message Z8 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message Z8. If format is invalid, send <EOT> or command-specific failure.

HOST	Dir.	PINPAD
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message ZA Set Transaction Display Context

Format: <STX>ZA[Data]<ETX>[LRC] (request frame)
 <STX>ZA[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Stores transaction amount, currency, and transaction type for the next card-read flow.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
ZA	2	Message ID
[Data]	Variable	[Type]<FS>[Amount]<FS>[Currency]<FS>[Symbol]
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
ZA	2	Message ID
[Status/Data]	Variable	Command-specific status, key code, text, or no frame depending on command.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message ZA request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message ZA. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message Z42 Read Key Code

Format: <STX>Z42[Data]<ETX>[LRC] (request frame)
 <STX>Z43[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Starts keypad single-key read after a prompt.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
Z42	3	Message ID
[Data]	Variable	[Timeout]
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
Z43	3	Message ID
[Status/Data]	Variable	Command-specific status, key code, text, or no frame depending on command.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message Z42 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Accept valid request and start the operation. Send the response when the operation completes. If successful, send Message Z43. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message Z50 String Entry

Format: <STX>Z50[Data]<ETX>[LRC] (request frame)
 <STX>Z51[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Starts keypad string entry after a prompt.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
Z50	3	Message ID
[Data]	Variable	[Entry control]
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
Z51	3	Message ID
[Status/Data]	Variable	Command-specific status, key code, text, or no frame depending on command.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message Z50 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Accept valid request and start the operation. Send the response when the operation completes. If successful, send Message Z51. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message Z43 Data Entry Response Identifier

Format: <STX>Z43[Data]<ETX>[LRC] (request frame)
<STX>Z43[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Device sends this after data entry completes.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description

<STX>	1	<02>
Z43	3	Message ID
[Payload]	Variable	Device-to-host response identifier.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
Z43	3	Message ID
[Payload]	Variable	Key code or entered text.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message Z43 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message Z43. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message Z51 Data Entry Response Identifier

Format: <STX>Z51[Data]<ETX>[LRC] (request frame)
 <STX>Z51[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Device sends this after data entry completes.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
Z51	3	Message ID
[Payload]	Variable	Device-to-host response identifier.
<ETX>	1	<03>

[LRC]	1	Checksum
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Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
Z51	3	Message ID
[Payload]	Variable	Key code or entered text.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message Z51 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message Z51. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Section 8 EMV Contact and Contactless Messages

Message T01 Load Contact Terminal Configuration

Format: <STX>T01[Data]<ETX>[LRC] (request frame)
 <STX>T02[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Stores terminal configuration data.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
T01	3	Message ID
[Payload]	Variable	Configuration packet or command-specific selector.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
T02	3	Message ID
[Status]	1	'0' success, '1' failure.
[Reason/Error]	Variable	Optional reason and error message.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message T01 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message T02. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message T03 Load Contact CAPK

Format: <STX>T03[Data]<ETX>[LRC] (request frame)
 <STX>T04[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Stores certification authority public key data.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
T03	3	Message ID
[Payload]	Variable	Configuration packet or command-specific selector.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
T04	3	Message ID
[Status]	1	'0' success, '1' failure.
[Reason/Error]	Variable	Optional reason and error message.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message T03 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message T04. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message T05 Load Contact Application Configuration

Format: <STX>T05[Data]<ETX>[LRC] (request frame)
<STX>T06[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Stores contact application configuration.

Message element:

Request frame (HOST to PINPAD)

Field	Length	Value and description
<STX>	1	<02>
T05	3	Message ID
[Payload]	Variable	Configuration packet or command-specific selector.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
T06	3	Message ID
[Status]	1	'0' success, '1' failure.
[Reason/Error]	Variable	Optional reason and error message.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message T05 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message T06. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message T07 Load EMV Data Format Table

Format: <STX>T07[Data]<ETX>[LRC] (request frame)
<STX>T08[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Stores tag format definitions for private and standard tags.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
T07	3	Message ID

[Payload]	Variable	Configuration packet or command-specific selector.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
T08	3	Message ID
[Status]	1	'0' success, '1' failure.
[Reason/Error]	Variable	Optional reason and error message.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message T07 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message T08. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message T09 Query Contact Configuration IDs

Format: <STX>T09[Data]<ETX>[LRC] (request frame)
<STX>T0A[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Returns stored contact configuration identifiers.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
T09	3	Message ID
[Payload]	Variable	Configuration packet or command-specific selector.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
T0A	3	Message ID
[Status]	1	'0' success, '1' failure.
[Reason/Error]	Variable	Optional reason and error message.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message T09 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message T0A. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message T0B Delete Contact Configuration

Format: <STX>T0B[Data]<ETX>[LRC] (request frame)

<STX>T0C[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Deletes stored contact configuration.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
T0B	3	Message ID
[Payload]	Variable	Configuration packet or command-specific selector.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description

<STX>	1	<02>
T0C	3	Message ID
[Status]	1	'0' success, '1' failure.
[Reason/Error]	Variable	Optional reason and error message.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message T0B request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message T0C. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message T51 Load Contactless Terminal Configuration

Format: <STX>T51[Data]<ETX>[LRC] (request frame)
<STX>T52[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Stores contactless terminal configuration.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
T51	3	Message ID
[Payload]	Variable	Configuration packet or command-specific selector.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
T52	3	Message ID
[Status]	1	'0' success, '1' failure.

[Reason/Error]	Variable	Optional reason and error message.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message T51 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message T52. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message T53 Load Contactless CAPK

Format: <STX>T53[Data]<ETX>[LRC] (request frame)
<STX>T54[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Stores contactless CAPK only when not already present.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
T53	3	Message ID
[Payload]	Variable	Configuration packet or command-specific selector.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
T54	3	Message ID
[Status]	1	'0' success, '1' failure.
[Reason/Error]	Variable	Optional reason and error message.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message T53 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message T54. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message T55 Load Contactless Application Configuration

Format: <STX>T55[Data]<ETX>[LRC] (request frame)
<STX>T56[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Stores contactless application configuration.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
T55	3	Message ID
[Payload]	Variable	Configuration packet or command-specific selector.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
T56	3	Message ID
[Status]	1	'0' success, '1' failure.
[Reason/Error]	Variable	Optional reason and error message.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
------	------	--------

HOST	Dir.	PINPAD
Message T55 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message T56. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message T59 Query Contactless Configuration IDs

Format: <STX>T59[Data]<ETX>[LRC] (request frame)
<STX>T5A[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Returns stored contactless configuration identifiers.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
T59	3	Message ID
[Payload]	Variable	Configuration packet or command-specific selector.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
T5A	3	Message ID
[Status]	1	'0' success, '1' failure.
[Reason/Error]	Variable	Optional reason and error message.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message T59 request frame	->	
	<-	<ACK>/<NAK>/<EOT>

HOST	Dir.	PINPAD
	<-	Validate command format and process request. If successful, send Message T5A. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message T5B Delete Contactless Configuration

Format: <STX>T5B[Data]<ETX>[LRC] (request frame)
 <STX>T5C[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Deletes stored contactless configuration.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
T5B	3	Message ID
[Payload]	Variable	Configuration packet or command-specific selector.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
T5C	3	Message ID
[Status]	1	'0' success, '1' failure.
[Reason/Error]	Variable	Optional reason and error message.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message T5B request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message T5C. If format is invalid, send <EOT> or command-specific failure.

HOST	Dir.	PINPAD
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message T5D Contactless Housekeeping

Format: <STX>T5D[Data]<ETX>[LRC] (request frame)
<STX>T5E[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Runs contactless housekeeping compatibility operation.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
T5D	3	Message ID
[Payload]	Variable	Configuration packet or command-specific selector.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
T5E	3	Message ID
[Status]	1	'0' success, '1' failure.
[Reason/Error]	Variable	Optional reason and error message.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message T5D request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message T5E. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>.	->	

HOST	Dir.	PINPAD
If canceling the session, send <EOT>.		

Message T5F Load Contactless DRL Configuration

Format: <STX>T5F[Data]<ETX>[LRC] (request frame)
 <STX>T5G[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Stores dynamic reader limit configuration.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
T5F	3	Message ID
[Payload]	Variable	Configuration packet or command-specific selector.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
T5G	3	Message ID
[Status]	1	'0' success, '1' failure.
[Reason/Error]	Variable	Optional reason and error message.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message T5F request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message T5G. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message T5H Delete Contactless DRL Configuration

Format: <STX>T5H[Data]<ETX>[LRC] (request frame)
 <STX>T5I[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Deletes dynamic reader limit configuration.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
T5H	3	Message ID
[Payload]	Variable	Configuration packet or command-specific selector.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
T5I	3	Message ID
[Status]	1	'0' success, '1' failure.
[Reason/Error]	Variable	Optional reason and error message.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message T5H request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message T5I. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message T11 Contact Application Selection

Format: <STX>T11[Data]<ETX>[LRC] (request frame)
 <STX>T12[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Starts contact application selection only. Multiple applications are shown to the user for touch or keypad selection.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
T11	3	Message ID
[Payload]	Variable	Transaction request, host response, DOL, or command-specific data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
T12	3	Message ID
[Status]	1	'0' success, '1' failure.
[Result/Data]	Variable	Transaction result code, reversal flag, TLV data, or reason/error fields.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message T11 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Accept valid request and start the operation. Send the response when the operation completes. If successful, send Message T12. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message T13 Contact Application Selection Alias

Format: <STX>T13[Data]<ETX>[LRC] (request frame)
<STX>T12[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Alias of contact application selection.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
T13	3	Message ID
[Payload]	Variable	Transaction request, host response, DOL, or command-specific data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
T12	3	Message ID
[Status]	1	'0' success, '1' failure.
[Result/Data]	Variable	Transaction result code, reversal flag, TLV data, or reason/error fields.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message T13 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Accept valid request and start the operation. Send the response when the operation completes. If successful, send Message T12. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message T15 Start Contact Transaction

Format: <STX>T15[Data]<ETX>[LRC] (request frame)
 <STX>T16[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Runs contact EMV transaction. Optional PIN scheme selector may be appended.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>

T15	3	Message ID
[Payload]	Variable	Transaction request, host response, DOL, or command-specific data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
T16	3	Message ID
[Status]	1	'0' success, '1' failure.
[Result/Data]	Variable	Transaction result code, reversal flag, TLV data, or reason/error fields.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message T15 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Accept valid request and start the operation. Send the response when the operation completes. If successful, send Message T16. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message T17 Complete Contact Online Authorization

Format: <STX>T17[Data]<ETX>[LRC] (request frame)
<STX>T16[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Completes online authorization and second card decision.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
T17	3	Message ID
[Payload]	Variable	Transaction request, host response, DOL, or command-specific data.

<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
T16	3	Message ID
[Status]	1	'0' success, '1' failure.
[Result/Data]	Variable	Transaction result code, reversal flag, TLV data, or reason/error fields.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message T17 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message T16. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message T19 Add Contact Issuer Script

Format: <STX>T19[Data]<ETX>[LRC] (request frame)
<STX>T20[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Stores issuer script data for completion.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
T19	3	Message ID
[Payload]	Variable	Transaction request, host response, DOL, or command-specific data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
T20	3	Message ID
[Status]	1	'0' success, '1' failure.
[Result/Data]	Variable	Transaction result code, reversal flag, TLV data, or reason/error fields.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message T19 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message T20. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message T1D Overwrite Contact Runtime Data

Format: <STX>T1D[Data]<ETX>[LRC] (request frame)
<STX>T1E[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Overwrites current transaction data objects.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
T1D	3	Message ID
[Payload]	Variable	Transaction request, host response, DOL, or command-specific data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>

T1E	3	Message ID
[Status]	1	'0' success, '1' failure.
[Result/Data]	Variable	Transaction result code, reversal flag, TLV data, or reason/error fields.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message T1D request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message T1E. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message T21 Query Contact Transaction Data

Format: <STX>T21[Data]<ETX>[LRC] (request frame)

<STX>T22[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Returns requested transaction tags.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
T21	3	Message ID
[Payload]	Variable	Transaction request, host response, DOL, or command-specific data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
T22	3	Message ID
[Status]	1	'0' success, '1' failure.
[Result/Data]	Variable	Transaction result code, reversal flag, TLV data, or reason/error fields.

<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message T21 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message T22. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message T25 Get Contact Batch Data

Format: <STX>T25[Data]<ETX>[LRC] (request frame)
<STX>T26[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Returns next batch data record.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
T25	3	Message ID
[Payload]	Variable	Transaction request, host response, DOL, or command-specific data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
T26	3	Message ID
[Status]	1	'0' success, '1' failure.
[Result/Data]	Variable	Transaction result code, reversal flag, TLV data, or reason/error fields.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message T25 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message T26. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message T27 Get Contact Online Authorization Data

Format: <STX>T27[Data]<ETX>[LRC] (request frame)
 <STX>T28[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Returns online authorization TLV data.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
T27	3	Message ID
[Payload]	Variable	Transaction request, host response, DOL, or command-specific data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
T28	3	Message ID
[Status]	1	'0' success, '1' failure.
[Result/Data]	Variable	Transaction result code, reversal flag, TLV data, or reason/error fields.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message T27 request frame	->	
	<-	<ACK>/<NAK>/<EOT>

HOST	Dir.	PINPAD
	<-	Validate command format and process request. If successful, send Message T28. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message T29 Get Contact Reversal Data

Format: <STX>T29[Data]<ETX>[LRC] (request frame)
 <STX>T2A[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Returns reversal TLV data.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
T29	3	Message ID
[Payload]	Variable	Transaction request, host response, DOL, or command-specific data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
T2A	3	Message ID
[Status]	1	'0' success, '1' failure.
[Result/Data]	Variable	Transaction result code, reversal flag, TLV data, or reason/error fields.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message T29 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message T2A. If format is invalid, send <EOT> or command-specific failure.

HOST	Dir.	PINPAD
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message T2B Get Contact Online Data Packet

Format: <STX>T2B[Data]<ETX>[LRC] (request frame)
<STX>T2C[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Returns online data in single final packet format.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
T2B	3	Message ID
[Payload]	Variable	Transaction request, host response, DOL, or command-specific data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
T2C	3	Message ID
[Status]	1	'0' success, '1' failure.
[Result/Data]	Variable	Transaction result code, reversal flag, TLV data, or reason/error fields.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message T2B request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message T2C. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>.	->	

HOST	Dir.	PINPAD
If canceling the session, send <EOT>.		

Message T2H Get Contact Transaction Data Packet

Format: <STX>T2H[Data]<ETX>[LRC] (request frame)
 <STX>T2I[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Returns online, batch, or reversal data in packet format.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
T2H	3	Message ID
[Payload]	Variable	Transaction request, host response, DOL, or command-specific data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
T2I	3	Message ID
[Status]	1	'0' success, '1' failure.
[Result/Data]	Variable	Transaction result code, reversal flag, TLV data, or reason/error fields.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message T2H request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message T2I. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message T31 Contact Card Data Read

Format: <STX>T31[Data]<ETX>[LRC] (request frame)
 <STX>T12[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Starts non-financial contact card data read.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
T31	3	Message ID
[Payload]	Variable	Transaction request, host response, DOL, or command-specific data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
T12	3	Message ID
[Status]	1	'0' success, '1' failure.
[Result/Data]	Variable	Transaction result code, reversal flag, TLV data, or reason/error fields.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message T31 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Accept valid request and start the operation. Send the response when the operation completes. If successful, send Message T12. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message T33 Fast Contact Read

Format: <STX>T33[Data]<ETX>[LRC] (request frame)
 <STX>T34[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Starts fast contact data/authorization read and returns packetized result.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
T33	3	Message ID
[Payload]	Variable	Transaction request, host response, DOL, or command-specific data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
T34	3	Message ID
[Status]	1	'0' success, '1' failure.
[Result/Data]	Variable	Transaction result code, reversal flag, TLV data, or reason/error fields.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message T33 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Accept valid request and start the operation. Send the response when the operation completes. If successful, send Message T34. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message T35 Non-EMV Contact Transaction

Format: <STX>T35[Data]<ETX>[LRC] (request frame)
<STX>T16[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Starts non-EMV contact transaction flow.

Message element:

Request frame (HOST to PINPAD)

Field	Length	Value and description
<STX>	1	<02>
T35	3	Message ID
[Payload]	Variable	Transaction request, host response, DOL, or command-specific data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
T16	3	Message ID
[Status]	1	'0' success, '1' failure.
[Result/Data]	Variable	Transaction result code, reversal flag, TLV data, or reason/error fields.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message T35 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Accept valid request and start the operation. Send the response when the operation completes. If successful, send Message T16. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message T37 Contact PIN Management

Format: <STX>T37[Data]<ETX>[LRC] (request frame)
<STX>T38[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Currently returns unsupported failure.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
T37	3	Message ID

[Payload]	Variable	Transaction request, host response, DOL, or command-specific data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
T38	3	Message ID
[Status]	1	'0' success, '1' failure.
[Result/Data]	Variable	Transaction result code, reversal flag, TLV data, or reason/error fields.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message T37 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message T38. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message T3C Force Complete Non-EMV Contact Flow

Format: <STX>T3C[Data]<ETX>[LRC] (request frame)
 <STX>T3C[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Completes non-EMV card-data flow.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
T3C	3	Message ID
[Payload]	Variable	Transaction request, host response, DOL, or command-specific data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
T3C	3	Message ID
[Status]	1	'0' success, '1' failure.
[Result/Data]	Variable	Transaction result code, reversal flag, TLV data, or reason/error fields.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message T3C request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message T3C. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message T61 Start Contactless Transaction

Format: <STX>T61[Data]<ETX>[LRC] (request frame)

<STX>T62[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Runs contactless EMV transaction. Optional PIN scheme selector may be appended.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
T61	3	Message ID
[Payload]	Variable	Transaction request, host response, DOL, or command-specific data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description

<STX>	1	<02>
T62	3	Message ID
[Status]	1	'0' success, '1' failure.
[Result/Data]	Variable	Transaction result code, reversal flag, TLV data, or reason/error fields.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message T61 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Accept valid request and start the operation. Send the response when the operation completes. If successful, send Message T62. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message T63 Query Contactless Transaction Data

Format: <STX>T63[Data]<ETX>[LRC] (request frame)
<STX>T64[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Returns requested contactless tags.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
T63	3	Message ID
[Payload]	Variable	Transaction request, host response, DOL, or command-specific data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
T64	3	Message ID
[Status]	1	'0' success, '1' failure.

[Result/Data]	Variable	Transaction result code, reversal flag, TLV data, or reason/error fields.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message T63 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message T64. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message T65 Query Contactless Online Data

Format: <STX>T65[Data]<ETX>[LRC] (request frame)
 <STX>T66[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Returns contactless online authorization data.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
T65	3	Message ID
[Payload]	Variable	Transaction request, host response, DOL, or command-specific data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
T66	3	Message ID
[Status]	1	'0' success, '1' failure.
[Result/Data]	Variable	Transaction result code, reversal flag, TLV data, or reason/error fields.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message T65 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message T66. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message T67 Query Contactless Payment Scheme

Format: <STX>T67[Data]<ETX>[LRC] (request frame)
<STX>T68[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Returns selected payment scheme code.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
T67	3	Message ID
[Payload]	Variable	Transaction request, host response, DOL, or command-specific data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
T68	3	Message ID
[Status]	1	'0' success, '1' failure.
[Result/Data]	Variable	Transaction result code, reversal flag, TLV data, or reason/error fields.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
------	------	--------

HOST	Dir.	PINPAD
Message T67 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message T68. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message T71 Complete Contactless Online Authorization

Format: <STX>T71[Data]<ETX>[LRC] (request frame)
<STX>T62[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Completes contactless online authorization.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
T71	3	Message ID
[Payload]	Variable	Transaction request, host response, DOL, or command-specific data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
T62	3	Message ID
[Status]	1	'0' success, '1' failure.
[Result/Data]	Variable	Transaction result code, reversal flag, TLV data, or reason/error fields.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message T71 request frame	->	
	<-	<ACK>/<NAK>/<EOT>

HOST	Dir.	PINPAD
	<-	Validate command format and process request. If successful, send Message T62. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message T73 Add Contactless Issuer Script

Format: <STX>T73[Data]<ETX>[LRC] (request frame)
 <STX>T74[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Stores contactless issuer script data.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
T73	3	Message ID
[Payload]	Variable	Transaction request, host response, DOL, or command-specific data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
T74	3	Message ID
[Status]	1	'0' success, '1' failure.
[Result/Data]	Variable	Transaction result code, reversal flag, TLV data, or reason/error fields.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message T73 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message T74. If format is invalid, send <EOT> or command-specific failure.

HOST	Dir.	PINPAD
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message T75 Load Contactless Revocation List

Format: <STX>T75[Data]<ETX>[LRC] (request frame)
<STX>T76[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Loads revocation list data.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
T75	3	Message ID
[Payload]	Variable	Transaction request, host response, DOL, or command-specific data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
T76	3	Message ID
[Status]	1	'0' success, '1' failure.
[Result/Data]	Variable	Transaction result code, reversal flag, TLV data, or reason/error fields.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message T75 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message T76. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>.	->	

HOST	Dir.	PINPAD
If canceling the session, send <EOT>.		

Message T77 Load Contactless Exception List

Format: <STX>T77[Data]<ETX>[LRC] (request frame)
 <STX>T78[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Loads exception list data.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
T77	3	Message ID
[Payload]	Variable	Transaction request, host response, DOL, or command-specific data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
T78	3	Message ID
[Status]	1	'0' success, '1' failure.
[Result/Data]	Variable	Transaction result code, reversal flag, TLV data, or reason/error fields.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message T77 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Validate command format and process request. If successful, send Message T78. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message T81 Start Contactless EMV Transaction

Format: <STX>T81[Data]<ETX>[LRC] (request frame)
 <STX>T62[Data]<ETX>[LRC] (response frame)

Message length: Variable.

Usage: Compatibility command to start contactless EMV transaction.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
T81	3	Message ID
[Payload]	Variable	Transaction request, host response, DOL, or command-specific data.
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<STX>	1	<02>
T62	3	Message ID
[Status]	1	'0' success, '1' failure.
[Result/Data]	Variable	Transaction result code, reversal flag, TLV data, or reason/error fields.
<ETX>	1	<03>
[LRC]	1	Checksum

Message flow:

HOST	Dir.	PINPAD
Message T81 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Accept valid request and start the operation. Send the response when the operation completes. If successful, send Message T62. If format is invalid, send <EOT> or command-specific failure.
Verify response frame. If valid, send <ACK>. If LRC error, send <NAK>. If canceling the session, send <EOT>.	->	

Message T1C Cancel Contact Transaction

Format: <STX>T1C[Data]<ETX>[LRC] (request frame)
 <ACK>/<NAK>/<EOT> only unless an error frame is specified.

Message length: Fixed.

Usage: Cancels active contact transaction.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
T1C	3	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<ACK>	1	Command accepted
<NAK>	1	Transmission error
<EOT>	1	Format error, timeout, or canceled session

Message flow:

HOST	Dir.	PINPAD
Message T1C request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Apply command action. No data frame is sent for a normal accepted command.

Message T23 Clear Contact Transaction Log

Format: <STX>T23[Data]<ETX>[LRC] (request frame)
<ACK>/<NAK>/<EOT> only unless an error frame is specified.

Message length: Fixed.

Usage: Clears transaction log state.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
T23	3	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<ACK>	1	Command accepted

<NAK>	1	Transmission error
<EOT>	1	Format error, timeout, or canceled session

Message flow:

HOST	Dir.	PINPAD
Message T23 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Apply command action. No data frame is sent for a normal accepted command.

Message T34 Fast Contact Packet Acknowledgement

Format: <STX>T34[Data]<ETX>[LRC] (request frame)
<ACK>/<NAK>/<EOT> only unless an error frame is specified.

Message length: Fixed.

Usage: Host compatibility acknowledgement for fast contact packets.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
T34	3	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<ACK>	1	Command accepted
<NAK>	1	Transmission error
<EOT>	1	Format error, timeout, or canceled session

Message flow:

HOST	Dir.	PINPAD
Message T34 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Apply command action. No data frame is sent for a normal accepted command.

Message T38 PIN Management Response Identifier

Format: <STX>T38[Data]<ETX>[LRC] (request frame)
<ACK>/<NAK>/<EOT> only unless an error frame is specified.

Message length: Fixed.

Usage: Compatibility response identifier.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
T38	3	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<ACK>	1	Command accepted
<NAK>	1	Transmission error
<EOT>	1	Format error, timeout, or canceled session

Message flow:

HOST	Dir.	PINPAD
Message T38 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Apply command action. No data frame is sent for a normal accepted command.

Message T6C Cancel Contactless Transaction

Format: <STX>T6C[Data]<ETX>[LRC] (request frame)
<ACK>/<NAK>/<EOT> only unless an error frame is specified.

Message length: Fixed.

Usage: Cancels active contactless transaction.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
T6C	3	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description

<ACK>	1	Command accepted
<NAK>	1	Transmission error
<EOT>	1	Format error, timeout, or canceled session

Message flow:

HOST	Dir.	PINPAD
Message T6C request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Apply command action. No data frame is sent for a normal accepted command.

Message T72 Contactless Compatibility Acknowledgement

Format: <STX>T72[Data]<ETX>[LRC] (request frame)
 <ACK>/<NAK>/<EOT> only unless an error frame is specified.

Message length: Fixed.

Usage: Accepted no-op used by some hosts.

Message element:

Request frame (HOST to PINPAD)		
Field	Length	Value and description
<STX>	1	<02>
T72	3	Message ID
<ETX>	1	<03>
[LRC]	1	Checksum

Message element:

Response frame (PINPAD to HOST)		
Field	Length	Value and description
<ACK>	1	Command accepted
<NAK>	1	Transmission error
<EOT>	1	Format error, timeout, or canceled session

Message flow:

HOST	Dir.	PINPAD
Message T72 request frame	->	
	<-	<ACK>/<NAK>/<EOT>
	<-	Apply command action. No data frame is sent for a normal accepted command.

Appendix A Fixed Prompt Identifiers

Prompt table:

Data entry prompts		
Field	Length	Value and description
001	3	ACCOUNT NUMBER
003	3	ENTER CUST ID
004	3	ENTER AMOUNT
015	3	ENTER CASH BACK
024	3	ENTER
100	3	ACCOUNT NUMBER
114	3	CUSTOMER REF
115	3	CUSTOMER REF NO.
126	3	ENTER BADGE #
129	3	ENTER CASH BACK
134	3	ENTER CUST REF
200	3	ENTER OTP
201	3	ENTER VERIFICATION CODE
202	3	ENTER CVV2
203	3	ENTER SMS OTP
204	3	ENTER AUTHENTICATOR CODE
205	3	ENTER TEMPORARY PASSWORD

Prompt table:

PIN prompts		
Field	Length	Value and description
001	3	ENTER PIN
002	3	REENTER PIN
006	3	ENTER NEW PIN
007	3	CONFIRM NEW PIN
024	3	ENTER